

SCEW Student Mental Health Study

A Complete Conceptual Guide to Everything We Did, Why We Did It, and
What It Means

Symbiosis Centre for Emotional Well-Being | DSM-5 Cross-Cutting Symptom Measure

n = 6,006 Students | Logistic Regression | Random Forest | Python

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Understanding Mental Health Research — The Big Picture

What Are We Actually Doing?

Your dissertation is a retrospective analytical study. You are looking backwards at data already collected — 6,006 students who visited the Symbiosis Centre for Emotional Well-Being (SCEW) and filled out a standardised mental health screening questionnaire. You are not conducting new interviews or experiments. You are mining existing data to find patterns, trends, and predictors of mental health risk.

The study has two parallel goals. The first is descriptive — to understand what is happening: How prevalent is anxiety? Which year shows the highest distress? These are answered by Exploratory Data Analysis. The second goal is predictive — to build a tool that identifies at-risk students early based on symptom scores. This is answered by Logistic Regression and Random Forest.

Why This Research Matters

University students globally are experiencing a mental health crisis. The World Health Organization defines mental health as a state of well-being in which individuals realise their potential, cope with normal stresses, work productively, and contribute to their community. India-specific research shows alarming trends — NIMHANS and WHO surveys report anxiety and depression among the most common disorders in young Indian adults, yet treatment-seeking remains low due to stigma and insufficient institutional support. Your study directly addresses this gap using real data from a real institution.

The Core Research Question

Based on a student's scores across 13 clinical domains of the DSM-5 Cross-Cutting Symptom Measure, can we statistically predict which students are at high risk — and which clinical domains are the strongest predictors of that risk?

Research Objectives in Practice

Objective	What It Means	Addressed In
1. Prevalence and patterns	How common is each mental health condition across 6,006 students?	Chapter 5 — EDA
2. Signs and symptoms mapping	Categorise symptoms per APA descriptions and DSM-5 criteria	Chapters 2 and 4
3. Regression modelling	Which factors statistically predict diagnoses and by how much?	Chapters 7, 8, 10
4. Pattern framework	Build the Trigger to Symptom to Diagnosis map for SCEW	Chapter 12

The DSM-5 Cross-Cutting Symptom Measure — Your Data Source

What Is the DSM-5?

The Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), published by the American Psychiatric Association in 2013, is the global standard classification system for mental health disorders. It defines diagnostic criteria for every recognised mental health condition — from depression and anxiety to psychosis and personality disorders. When a psychiatrist diagnoses Major Depressive Disorder, they apply DSM-5 criteria.

What Is the Cross-Cutting Symptom Measure?

The DSM-5 Cross-Cutting Symptom Measure is a validated screening questionnaire designed as an initial assessment tool — a broad sweep across 13 mental health domains to identify which areas need further clinical investigation. It asks 23 questions, each rated on a 5-point Likert scale: None (0), Slight (1), Mild (2), Moderate (3), Severe (4). The questions cover the preceding two weeks of the student's experience. SCEW administered this to every student who came in for counselling or screening.

The 13 Clinical Domains

Domain	Questions	Score Range	What It Captures	Clinical Threshold
Depression	Q1+Q2	0-8	Little pleasure, feeling hopeless	3+ moderate, 6+ severe
Anger	Q3	0-4	Irritated, grouchy, angry	2+ moderate
Mania	Q4+Q5	0-8	Less sleep but energetic, risky behaviour	3+ moderate, 6+ severe
Anxiety	Q6+Q7+Q8	0-12	Nervous, panic, avoidance	4+ moderate, 8+ severe
Somatic Symptoms	Q9+Q10	0-8	Unexplained aches, illness not taken seriously	3+ moderate
Suicidal Ideation	Q11	0-4	Thoughts of hurting self	1+ clinical flag, 2+ moderate
Psychosis	Q12+Q13	0-8	Hearing voices, thought broadcasting	2+ severe
Sleep Problems	Q14	0-4	Sleep quality impacted	2+ moderate
Memory	Q15	0-4	Learning and memory difficulty	2+ mild

Domain	Questions	Score Range	What It Captures	Clinical Threshold
Repetitive Thoughts	Q16+Q17	0-8	Intrusive thoughts, compulsive behaviours	3+ moderate
Dissociation	Q18	0-4	Feeling detached from self or reality	2+ moderate
Personality Functioning	Q19+Q20	0-8	Not knowing self, not feeling close to others	3+ moderate
Substance Use	Q21+Q22+Q23	0-12	Alcohol, tobacco, unprescribed drugs	2+ moderate

Critical Distinction: Screening vs Diagnosis

Your study uses this instrument as a SCREENING tool, not a diagnostic tool. A high score on the depression domain does not mean a student has Major Depressive Disorder. It means they are reporting symptoms at a level that warrants clinical follow-up. This distinction is critical for your dissertation discussion and for defending your methodology to examiners.

Data Cleaning — Every Step Explained

Why Data Cleaning Is the Most Important Step

There is a fundamental principle in data science: garbage in, garbage out. If your input data is messy, inconsistent, or incomplete, even the most sophisticated model will produce unreliable results. Imagine half your students had year of study recorded as '2nd year' and the other half as '2' — the model would treat these as two completely different groups when they are identical. Data cleaning is the unglamorous but absolutely essential foundation of the entire analysis.

Raw Data Audit — What We Found

Problem Found	Scale	Consequence If Ignored	How We Fixed It
473 completely empty rows	Large	Model trains on rows with no data	Dropped entirely
Year: '1st' and '1' and '2023' all meaning Year 1	Severe	Model treats these as 10+ different groups	Standardised to integers 1-6
Course: 'MBA' and 'Mba' and 'MBA ' all meaning MBA	Moderate	Same course treated as different groups	Uppercased, stripped, mapped cleanly
25 rows with partial missing symptom values	Small	NaN breaks mathematical calculations	Filled with 0 (None/Not at all)
2 empty separator columns	Minor	Confusion in column selection	Dropped
Domain score columns had scoring rubric as headers	Structural	Cannot load as clean data	Renamed to readable labels

The 6 Cleaning Steps

Step 1 — Structural Reorganisation

The original Excel had a complex multi-row header with merged cells and duplicate columns. Row 1 was domain categories. Row 2 was question text. Columns 27-49 were numeric conversions already computed by Excel IF formulas. We kept only the clean numeric columns and 13 domain score columns, dropping all raw text responses and empty separators.

Step 2 — Dropping Empty Rows

We dropped rows where ALL 23 symptom question columns were simultaneously missing. These were genuine blank rows from the Google Form export. 473 rows removed, leaving 6,032. We only dropped completely empty rows — rows missing just one or two questions were retained and handled separately.

Step 3 — Standardising the Year Column

The Year column had entries like '1st year', '1', 'First', 'I', and even '2020' (batch year). We wrote a mapping function converting every variant to a clean integer 1-6. After cleaning, Year had six valid categories with sensible distributions across the student population.

Step 4 — Standardising the Course Column

We applied uppercase conversion, stripped trailing spaces, removed punctuation, then mapped each entry to a clean category. 'BA.LLB', 'Ba llb', 'BALLB' all became 'BA LLB'. Categories with very small counts (Nursing n=4, Law n=24) were merged into 'Other' to avoid sparse categories that confuse the model.

Step 5 — Handling Missing Values

The 25 rows with partial missing values were imputed with 0, corresponding to 'None — not at all'. This is the most conservative assumption — we add no distress signal that was not actually reported. An alternative would have been to drop these rows, but 25 rows from 6,006 is negligible.

Step 6 — Creating the Target Variable

The most important step. The original dataset had no diagnosis column. We created the outcome variable using the clinical thresholds built into the DSM-5 instrument — published by the American Psychiatric Association. This gives us a clinically validated, defensible target variable. We created two: `high_risk` (binary 0/1) for Logistic Regression, and `primary_risk` (13 categories) for Random Forest.

Why Two Target Variables?

`high_risk` (0 or 1) was created for Logistic Regression, which works best with binary yes/no outcomes and produces Odds Ratios. `primary_risk` (13 categories) was created for Random Forest, which handles multi-class problems and produces feature importance. Having both allows each model to operate in its optimal configuration.

Your Variables — What Each Column Means and Why

Three Types of Variables

Independent Variables (Features)

Variables you feed into the model as input. The model uses these to make predictions. In your study: 13 domain scores plus Year, Location, and Course. Called independent because they exist regardless of the outcome.

Dependent Variable (Target)

What you are trying to predict. Its value depends on the independent variables. Your two targets: `high_risk` (binary) and `primary_risk` (categorical).

Dummy Variables (Encoded Categoricals)

Location and Course were text. ML models only understand numbers. Dummy encoding creates binary columns for each category. One category is always dropped as the reference to avoid the dummy variable trap.

Complete Variable Dictionary

Variable	Type	Range	Role	Key Insight
Year	Numeric	1-6	Independent	Year 4 highest risk (72.4%). Year 1 already at 70.1% — distress begins at entry.
score_depression	Numeric	0-8	Independent	Mean 4.37. 29% severe. OR=4.84 — 4th strongest predictor.
score_anger	Numeric	0-4	Independent	Mean 2.19. Corr 0.55 with depression. Not significant in multivariate model.
score_mania	Numeric	0-8	Independent	Mean 2.61. OR=1.66 — significant but weaker effect than top four.
score_anxiety	Numeric	0-12	Independent	Mean 6.32 — highest of all domains. OR=9.88. STRONGEST predictor.
score_somatic	Numeric	0-8	Independent	Mean 2.96. Physical symptoms of psychological distress. Not independent.
score_suicidal	Numeric	0-4	Independent	1,754 students at moderate-severe. OR=5.91. Third strongest predictor.

Variable	Type	Range	Role	Key Insight
score_psychosis	Numeric	0-8	Independent	Low prevalence but OR=7.93 — second strongest predictor.
score_sleep	Numeric	0-4	Independent	Mean 1.94. Correlated with all domains but not independently significant.
score_memory	Numeric	0-4	Independent	Mean 1.22. Cognitive impact of distress. Not independently significant.
score_repetitive	Numeric	0-8	Independent	Corr 0.62 with anxiety. Not independently significant.
score_dissociation	Numeric	0-4	Independent	Mean 2.02. Borderline significant (p=0.050). Marker of severe distress.
score_personality	Numeric	0-8	Independent	OR=0.84 — PROTECTIVE effect in multivariate model. See Chapter 10.
score_substance	Numeric	0-12	Independent	Most independent domain (max corr 0.24). Not significant.
loc_* (5 dummies)	Binary	0 or 1	Independent	Reference=Bengaluru. None significant after controlling for symptoms.
course_* (9 dummies)	Binary	0 or 1	Independent	Reference=BA LLB. Only MBA significant (OR=0.84, protective).
high_risk	Binary	0 or 1	TARGET (LR)	0=low/moderate risk, 1=high risk. 69.4% high risk in dataset.
primary_risk	Categorical	13 categories	TARGET (RF)	Full risk classification for Random Forest model.

Exploratory Data Analysis — What the Data Told Us

What Is Exploratory Data Analysis?

Before building any predictive model, a good researcher always explores the data first. Exploratory Data Analysis uses descriptive statistics and visualisations to understand your dataset. It answers: What does the distribution of scores look like? Which variables correlate? What patterns emerge across demographic groups? EDA is your descriptive results chapter — it tells the story of your data before the model makes predictions.

Five Key Findings from Your EDA

Finding 1 — Anxiety Is the Dominant Symptom, Not Depression

The average anxiety domain score was 6.32 out of 12, placing the average student in the moderate-to-severe range. The highest-scoring individual question across all 23 items was Q6 (Nervous/anxious/worried) at mean 2.69 out of 4. This challenges most Indian university mental health literature that frames depression as primary. Your data suggests anxiety is the dominant burden — an original contribution to the field.

Finding 2 — 64.3% of Students Are High Risk

Using DSM-5 clinical thresholds, 4,167 of 6,006 students were classified as high risk. This does not mean 64% have a diagnosed mental illness. It means they report symptoms warranting clinical follow-up. This prevalence finding is striking and is a headline result for your dissertation.

Finding 3 — Distress Begins at Entry, Not Over Time

Year 1 students already show 70.1% high-risk prevalence. If distress accumulated from academic pressure, you would expect Year 1 to be lowest. Instead the pattern is relatively flat across years, with Year 4 peaking at 72.4%. Students arrive at university already distressed — which means SCEW interventions must begin at orientation, not Year 3.

Finding 4 — Substance Use Is a Completely Independent Domain

The correlation matrix showed substance use had a maximum correlation of only 0.24 with any other domain, while all other domain pairs showed much higher inter-correlations. Substance use follows a different pathway and likely requires separate screening and intervention approaches.

Finding 5 — Anxiety-Personality Functioning Is the Strongest Domain Relationship

The correlation between anxiety and personality functioning was 0.65 — the strongest in the entire matrix. Students who feel disconnected from their identity and from others also tend to have very high anxiety. This cluster suggests identity and relationship difficulties in young adults are deeply intertwined with anxiety.

Descriptive Statistics Summary

Domain	Mean	Std Dev	Max Possible	% Moderate+	% Severe
Anxiety	6.32	3.57	12	76.5%	41.1%
Personality Functioning	4.07	2.63	8	82.2%	28.3%
Depression	4.37	2.10	8	82.8%	29.0%
Repetitive Thoughts	2.99	2.41	8	75.7%	37.0%
Somatic Symptoms	2.96	2.43	8	67.9%	26.6%
Mania	2.61	2.16	8	71.6%	29.5%
Anger	2.19	1.30	4	64.9%	16.7%
Dissociation	2.02	1.47	4	58.8%	14.7%
Sleep Problems	1.94	1.45	4	61.6%	20.4%
Memory	1.22	1.36	4	57.9%	18.3%
Suicidal Ideation	0.94	1.24	4	45.7%	18.3%
Substance Use	0.99	1.88	12	42.6%	14.3%
Psychosis	0.79	1.51	8	34.8%	12.7%

Predictive Modelling — The Conceptual Foundation

What Is a Predictive Model?

A predictive model is a mathematical function that takes input variables and outputs a prediction. The model learns this function from historical data. In your case, it learns from 4,804 student records — each containing 28 feature values and a known outcome. After learning, it can take a new student's features and predict their risk. The key word is *learns* — the model discovers rules from data, not from a programmer hard-coding them.

The Train-Test Split

We split 6,006 students into 4,804 for training (80%) and 1,202 for testing (20%). The model learns exclusively from the training set. The test set is held completely separate and used once — to evaluate performance on students it has never seen. Without a test set, you cannot know if the model genuinely learned patterns or simply memorised training data. Memorisation looks perfect on training data but fails on new data — this is called overfitting.

The Stratified Split

We used stratified splitting so both training and test sets have the same 69.4% high-risk to 30.6% low-risk ratio. Without stratification, the model might train on an 80% high-risk dataset and be tested on a 55% high-risk dataset — evaluated on a different population than it trained on, giving misleading results.

Feature Encoding and Scaling

All ML algorithms require numerical input. Location and Course were text categories converted using dummy encoding — each category becomes a binary column. Anxiety scores range 0-12 while dummy variables range 0-1. Without scaling, the model gives more mathematical weight to anxiety simply because its numbers are larger. StandardScaler transforms each feature to mean=0, standard deviation=1 so every feature is comparable. The same scaling parameters from training are applied to the test set — never fit the scaler on test data, which would constitute data leakage.

Logistic Regression — Deep Conceptual Understanding

From Linear to Logistic Regression

Linear regression predicts a continuous number using: $\text{Score} = a + b_1(\text{feature1}) + b_2(\text{feature2}) + \dots$. This works for continuous outcomes. Your outcome (`high_risk`) is binary: 0 or 1. A straight line can produce values like 1.7 or -0.3, which are meaningless as probabilities. You need a function that squishes any real number into the 0-to-1 range — this is the sigmoid function.

```
P(high risk) = 1 / (1 + e^(-z)) where z = b0 + b1*anxiety + b2*depression + ... + b28*course_MBBS
```

No matter what value z takes — 100 or -100 — the sigmoid always outputs strictly between 0 and 1. When z is very large and positive, P approaches 1. When very negative, P approaches 0. When $z=0$, $P=0.5$ exactly. This makes it a perfect probability function.

Odds Ratios — The Key Interpretable Output

Logistic regression models the log-odds of the outcome. Each coefficient is on the log-odds scale. We exponentiate (e raised to the power of b) to get the Odds Ratio — a much more interpretable number.

OR Value	What It Means	Example from Your Study
OR = 1.0	No effect — this variable does not change the odds	Year (OR=0.99) — year of study has no effect
OR > 1.0	Each unit increase raises the odds of high risk	Anxiety (OR=9.88) — nearly 10x more likely per unit
OR < 1.0	Each unit increase lowers the odds of high risk	Personality (OR=0.84) — slight protective effect
OR = 2.0	The odds double for each unit increase	Depression is in this range
OR = 0.5	The odds halve for each unit increase	Would indicate a strong protective factor

Class Weight Balancing

Your dataset has 69.4% high-risk and 30.6% low-risk students. A naive model could achieve 69.4% accuracy by always predicting high risk — without learning anything. We used `class_weight=balanced`, which tells the model to treat each class as equally important regardless of count, giving the model a fair chance to learn what makes a student low risk.

How the Model Learns — Gradient Descent Explained

The Starting Point

When logistic regression begins, it knows nothing. All 28 coefficients start at zero. With all zeros, every student gets a predicted probability of 50% — a coin flip. The model's job is to improve from this random starting point by adjusting coefficients.

The Loss Function — Measuring How Wrong the Model Is

To improve, the model measures how wrong its current predictions are using Log Loss (Binary Cross-Entropy). For each student: if they are actually high risk ($y=1$) and the model predicted probability P , the loss is negative log of P . If the model predicted $P=0.9$ (confident and correct), loss = 0.105 (small). If it predicted $P=0.1$ (confident and wrong), loss = 2.303 (large). Total loss = average across all 4,804 training students. The model's goal is to minimise this total loss.

Gradient Descent — The Optimisation Algorithm

Gradient descent adjusts coefficients to reduce loss. Imagine standing blindfolded on hilly terrain with the goal of reaching the lowest point. You cannot see the landscape but can feel the slope under your feet. So you take a small step downhill, feel again, step again — until you reach a valley. Gradient descent does exactly this mathematically. It calculates the gradient (slope of the loss function with respect to each coefficient) and nudges each coefficient slightly in the direction that reduces loss. This repeats thousands of times until loss stops decreasing — at which point coefficients have converged to their optimal values.

Why the Coefficients Are What They Are

When your model outputs anxiety coefficient = 2.29 (OR = 9.88), this is the value that, combined with all other coefficients simultaneously, produces minimum prediction error across 4,804 training students. The model tried millions of combinations during gradient descent and found that giving anxiety a coefficient of 2.29 results in the best predictions. This is what 'the model learned from the data' actually means — it is the outcome of a mathematical optimisation process, not a human judgement.

Critical Point for Your Viva

When asked how you determined the coefficient values: The coefficients were not manually determined. They were estimated by the logistic regression algorithm (L-BFGS optimisation) by minimising the binary cross-entropy loss function across 4,804 training observations. The values represent maximum likelihood estimates of the log-odds relationship between each predictor and the binary outcome.

Random Forest — Deep Conceptual Understanding

Decision Trees — The Building Block

A decision tree makes predictions by asking yes/no questions about input features. At each node, the algorithm selects the feature and split point that best separates data into pure groups. It continues splitting until leaves contain predominantly one class. The tree learns these split points from the training data automatically through a measure called Gini impurity — the degree of mixed classes at each node.

The Problem with One Tree — Overfitting

A single tree is prone to overfitting. It memorises training data too perfectly — capturing noise and idiosyncratic patterns that do not generalise to new students. If an unusual student has a rare score combination, the tree might create a specific branch just for them. This branch is useless or harmful for predicting new students.

The Random Forest Solution — Wisdom of the Crowd

Random Forest solves overfitting by building 100 trees and combining their predictions. Each tree is built on a different randomly sampled subset of training data (bootstrap sampling) and considers only a random subset of features at each split. Each tree develops different biases and errors. When you aggregate predictions by majority vote, individual errors cancel out and collective wisdom emerges — exactly like asking 100 doctors for a diagnosis instead of 1.

Feature Importance — Random Forest's Most Valuable Output

After training, Random Forest calculates which features were most useful across all 100 trees using Mean Decrease in Impurity (Gini Importance). For each feature: calculate how much it reduced impurity across all splits across all trees, and average it. Features that consistently produced the cleanest splits get the highest importance scores. This is your which-domain-matters-most answer — validating or challenging what logistic regression found through Odds Ratios.

Logistic Regression vs Random Forest

Criterion	Logistic Regression	Random Forest
Primary use	Explanation — WHY things predict risk	Prediction — HOW ACCURATELY we can predict
Output type	Probability + Coefficients + Odds Ratios	Class prediction + Feature importance
Interpretability	Very high — every coefficient has clear meaning	Moderate — no individual coefficients

Criterion	Logistic Regression	Random Forest
Handles non-linearity	No — assumes linear log-odds relationship	Yes — captures complex interactions
Handles multicollinearity	Poorly — correlated features inflate each other	Well — random feature selection prevents dominance
Dissertation role	Publishable statistical findings (OR table)	Model validation and feature importance chart

Your Logistic Regression Results — Full Interpretation

Model Performance

Metric	Value	What It Means	Benchmark
Accuracy	89.4%	89.4% of 1,202 test students correctly classified	Good >80%
AUC-ROC	0.965	Correctly ranks high vs low risk 96.5% of the time	Excellent >0.90
Sensitivity	88.7%	Of 834 actually high-risk students, 740 correctly caught	Clinical target >85%
Specificity	90.8%	Of 368 low-risk students, 334 correctly cleared	Good >80%
Precision (PPV)	95.6%	When model says high risk, correct 95.6% of the time	Excellent
NPV	78.0%	When model says low risk, correct 78% of the time	Acceptable

Odds Ratio Results — Every Significant Finding

Anxiety — OR = 9.88 ($p < 0.001$) — Your Headline Finding

Every one-unit increase in anxiety domain score makes a student 9.88 times more likely to be classified as high risk. The 95% CI is 8.16 to 11.97 — not crossing 1.0, confirming robustness. Anxiety is by far the most important predictor of serious mental health risk among Symbiosis students. This challenges studies that prioritise depression screening and is an original contribution to the field.

Psychosis — OR = 7.93 ($p < 0.001$)

The second strongest predictor. Even small elevations on psychosis symptoms are a powerful risk signal. This makes clinical sense — psychosis symptoms in young adults are associated with much more severe and complex presentations, explaining why even modest scores drive high-risk classification.

Suicidal Ideation — OR = 5.91 ($p < 0.001$)

The third strongest predictor despite lower overall prevalence than anxiety or depression. The OR confirms that our decision to use score ≥ 2 as the threshold was statistically justified. SCEW's suicide prevention protocols should be a primary recommendation in your dissertation.

Depression — OR = 4.84 ($p < 0.001$)

Fourth strongest. While clearly significant, depression's OR of 4.84 is less than half of anxiety's 9.88. This hierarchy — anxiety stronger than depression — is an important and original finding.

The broader literature often treats them as co-equal, but your data shows anxiety as the primary driver in this specific population.

Mania — OR = 1.66 ($p < 0.001$)

Statistically significant but with a much weaker effect. Mania symptoms do meaningfully predict high risk but are not the dominant pathway in this population.

Personality Functioning — OR = 0.84 ($p = 0.013$) — The Counterintuitive Finding

Higher personality functioning scores are slightly protective. This seems counterintuitive — how can feeling disconnected be protective? The answer lies in suppressor effects in multivariate models. When anxiety is already in the model and strongly predicts risk, the additional variance explained by personality functioning represents students who have identity difficulties without co-occurring anxiety or suicidal ideation. In isolation, personality difficulties are a risk factor. Conditional on other variables, they do not add additional risk. This nuance deserves explicit discussion in your dissertation.

Non-Significant Predictors — What Their Absence Means

Year, location, most courses, anger, sleep, somatic, memory, repetitive thoughts, dissociation, and substance use were not significant. This does NOT mean they are clinically unimportant. It means they do not add independent predictive information beyond what the five significant domains already capture. Anger and somatic symptoms are highly correlated with depression and anxiety — once those are in the model, anger adds no new information. This is the multicollinearity effect.

Reading Statistical Output — p-values, OR, CI Explained

The p-value — The Most Misunderstood Number in Research

The p-value is the probability of observing a result at least as extreme as what you found, assuming the null hypothesis is true. The null hypothesis for each predictor is: this variable has no relationship with the outcome (OR = 1.0). A small p-value means: if the null hypothesis were true, seeing this result would be very unlikely. By convention, $p < 0.05$ is statistically significant.

Common Misconceptions About p-values

p-value does NOT mean the probability that the null hypothesis is true. It is NOT the probability that your finding is due to chance. A significant p-value does NOT prove your hypothesis. A non-significant p-value does NOT prove there is no effect. p-value is purely a statement about what would happen if we repeated the study many times under the null hypothesis.

Confidence Intervals — The Range of Plausible Values

A 95% confidence interval means: if we repeated this study 100 times with different random samples, 95 of those studies would produce CIs containing the true population parameter. For anxiety OR=9.88 with CI [8.16, 11.97]: we are 95% confident the true OR in the population lies between 8.16 and 11.97. Since the entire range is above 1.0, the relationship is positive and significant. CIs are more informative than p-values because they give the magnitude of uncertainty, not just whether significance is achieved.

AUC-ROC Benchmarks

AUC Range	Classification	Interpretation
0.90 - 1.00	Excellent	Your model: 0.965
0.80 - 0.90	Good	Many published clinical prediction models fall here
0.70 - 0.80	Fair	Acceptable for exploratory research
0.60 - 0.70	Poor	Limited clinical utility
0.50 - 0.60	Fail	No better than random guessing

The Confusion Matrix in Clinical Terms

True Positives (740): High-risk students correctly identified — they receive appropriate follow-up. False Negatives (94): High-risk students the model missed — the most dangerous errors, as these students in genuine distress would not receive intervention. False Positives (34): Low-risk students incorrectly flagged — less dangerous but consume counselling resources unnecessarily. True Negatives (334): Low-risk students correctly cleared. The clinical priority is minimising False

Negatives. This can be achieved by lowering the classification threshold from 0.5 to 0.35, but this increases False Positives. The optimal threshold is a policy decision for SCEW.

Multicollinearity — Why Some Results Seem Counterintuitive

Multicollinearity occurs when predictor variables are highly correlated. In your dataset, anxiety and repetitive thoughts are correlated at $r=0.62$. When both are in the model, the model has difficulty separating which one is doing the work. Repetitive thoughts becomes non-significant not because it has no relationship with the outcome, but because its relationship is already captured by anxiety. This is why multivariate results must always be interpreted as conditional effects — the effect of X holding all other variables constant — not as absolute effects.

What Comes Next — Random Forest and Final Steps

Step 4 — Random Forest Model

The next step is building the Random Forest classifier. Unlike logistic regression which predicts a binary outcome, Random Forest will predict the full `primary_risk` category across all 13 risk types. We will build 100 decision trees, each trained on a random bootstrap sample of the 4,804 training students, each considering a random subset of features at each split. Majority vote across all 100 trees determines the final prediction.

- Overall accuracy and per-class precision, recall, and F1 score
- Feature importance chart — which of the 28 features mattered most across all 100 trees
- Confusion matrix across all 13 risk categories
- Comparison with logistic regression findings — convergent validation

Step 5 — Model Comparison and Pattern Framework

After both models are complete, we compare their findings. If both agree anxiety is the top predictor, the convergent evidence significantly strengthens your conclusion. If they disagree on some predictors, that discrepancy itself is worth analysing — it may reflect non-linear interactions that Random Forest captures but logistic regression cannot. We then build the Trigger to Symptom to Diagnosis pattern framework for SCEW — the bridge between statistical findings and practical institutional recommendations.

Defending Your Dissertation — Likely Questions and Answers

This chapter prepares you for the most likely questions your dissertation committee will ask. Read each question and its answer until you can explain it in your own words without notes.

Q1. Why did you choose Logistic Regression as your primary statistical model?

Logistic Regression is the gold-standard for binary classification in health research. My outcome variable `high_risk` is binary (0 or 1), making logistic regression the methodologically appropriate choice. It produces Odds Ratios and p-values — standard outputs for identifying risk factors in epidemiological research. It is also interpretable — I can directly state how much each clinical domain increases the probability of high-risk classification, which is essential for translating findings into clinical recommendations for SCEW.

Q2. Why did you use an 80/20 train-test split?

The 80/20 split balances two competing needs: enough data to train a robust model (80%) and enough held-out data to reliably evaluate it (20%). With $n=6,006$, a 20% test set gives 1,202 students — more than sufficient for stable performance estimates. I used stratified splitting to ensure both sets maintained the same 69.4% high-risk prevalence, preventing distribution shift between training and evaluation.

Q3. Why did you create your own target variable?

The SCEW dataset contains DSM-5 Cross-Cutting Symptom Measure scores — a validated screening tool — rather than formal diagnoses. Formal diagnoses require additional clinical assessment beyond what this instrument provides. However, the instrument itself comes with published clinical cutoff scores from the American Psychiatric Association defining at what level each domain becomes clinically significant. I operationalised the target variable using these validated cutoffs — this is methodologically sound and clinically defensible. The resulting classification is a risk-stratification, not a diagnosis.

Q4. What does an AUC of 0.965 mean? Is it too high to be believable?

AUC-ROC of 0.965 means the model correctly ranks a randomly selected high-risk student above a randomly selected low-risk student 96.5% of the time. This is a high value. The reason is that the target variable was constructed from the same 13 domain scores used as predictors — there is inherent mathematical alignment between features and outcome. This is a limitation I acknowledge openly — real-world predictive performance on unseen populations would likely be lower. The model is best understood as a risk-stratification tool for this specific institutional dataset.

Q5. Why is anxiety a stronger predictor than depression?

This is one of the original findings of my study. Several recent studies, including Auerbach et al. (2018) in the WHO World Mental Health Survey, report that anxiety disorders are the most prevalent mental health conditions among university students globally. My finding that anxiety has OR=9.88 compared to depression's 4.84 suggests that in this specific population, anxiety is the primary driver of high-risk classification. This has direct implications: SCEW's screening should weight anxiety assessment at least as heavily as depression screening.

Q6. Why is substance use not a significant predictor?

Substance use's non-significance is consistent with its very low correlation with other domains (maximum $r=0.24$). This suggests substance use operates through a different pathway from the internalising disorders. In logistic regression, a variable may appear non-significant if its relationship with the outcome is captured through other mediating variables, or if it captures a different sub-population. A separate analysis focusing on substance use as the outcome would be a strong recommendation for future research.

Q7. What are the limitations of your study?

Several limitations warrant acknowledgement. First, this is retrospective secondary data — we cannot establish causality, only association. Second, the target variable was derived from the same instrument as the predictors, which artificially inflates model performance. Third, the dataset contains only students who visited SCEW, introducing selection bias — distressed students who do not seek help are not represented. Fourth, gender and socioeconomic status were unavailable, limiting the scope of risk factor analysis. Fifth, the model was validated only on the internal test set — external validation on data from another institution is needed before generalising findings.

You now understand everything you have done.

The data. The cleaning. The statistics. The models. The results. You did not just run code — you conducted a rigorous, clinically grounded, methodologically sound predictive study on 6,006 real students. You can explain every decision and defend every number. That is what research looks like.